Challenge

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Description

Scenario

The NED 2 Robotic Arm makes a cup of coffee without human intervention.
The student discovers the basic functions of the Ned2 robotic arm. In this exercise,
the Ned2 robotic arm, using the basic blocks of the Blockly “No Code” programming environment, performs a simple pick and place operation with a cup of coffee and pressing a button on the coffee machine.
The exercise aims to show the student how to configure a robotics cell, how to adapt a gripper to specific needs (size of the coffee cup and button of the coffee machine) and how to carry out the sequence of movements to achieve the stain.
This exercise lends itself to a competition between different groups of students

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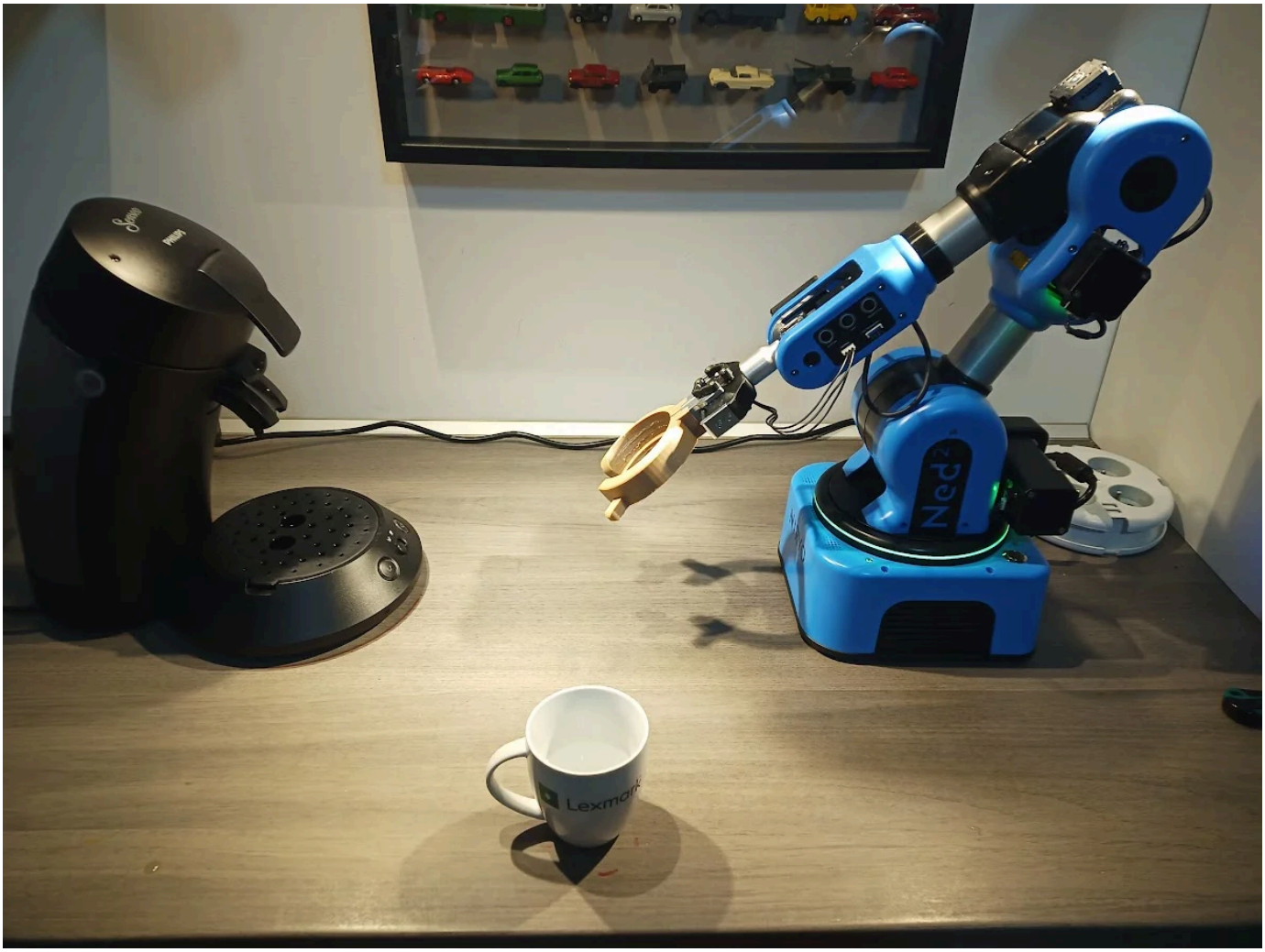
4.5. Final Solution

1. Chapter 1: Creation of the robotic cell

Objective

Optimize the placement of the robotic arm, coffee machine and cup

You've probably seen a video on Youtube of a robot preparing coffee. The task seems simple, but it requires careful preparation and a lot of thought to optimize each movement. Picking up a cup of coffee, especially if it's irregularly shaped, requires skill and precision. If we add to this the fact that with the same gripper we have to perform two different tasks, moving a cup and pressing buttons, the task requires an additional effort of creativity.
This exercise does not have a single solution as any coffee machine and any cup can be used, as can any size table, but overall what is asked of the student is that the NED2 robotic arm picks up a cup placed in a specific place on the table, positions it on the coffee machine, presses the button for the machine to make the coffee, waits for the coffee to be made and, once finished, the NED2 robotic arm must return the cup to the original location.
First, position the NED2 robotic arm, the coffee machine and the cup on the table in a position that seems logical to you, in which you consider that the NED2 robotic arm will be able to perform the required movements comfortably. If you wish, consult the "Discovery of the 6 axes of the NED2 Robotic Arm" to review concepts such as "the work envelope" or "singularities" to help you in your decision to position the NED2 robotic arm, the cup and coffee machine.



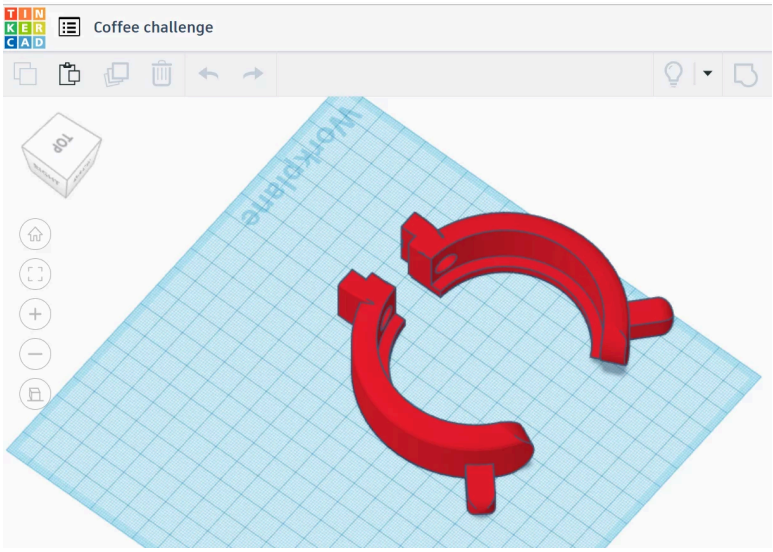
2. Chapter 2: Modifying the “Custom Gripper”

Objective

Create a tool specific to the task to be carried out

> **Note**
>
> This step can be skipped if using a small cup that can be gripped by the adaptive gripper.

We are faced with the problem of delicately and very precisely grasping an object in order to prevent the coffee from spilling.
For this, in the example in the image below, Tinkercad was used to design a clamp that fits the cup size perfectly and also has protrusions for pressing buttons.
On this link (STL file), you will find the 3D model of the clamp with which the “Custom Gripper” is equipped and the models of this example.
From these models, and using any 3D modeling program, design the clamp that fit the cup you are going to use. Remember that with the same clamp you will also have to press the buttons on the coffee machine.




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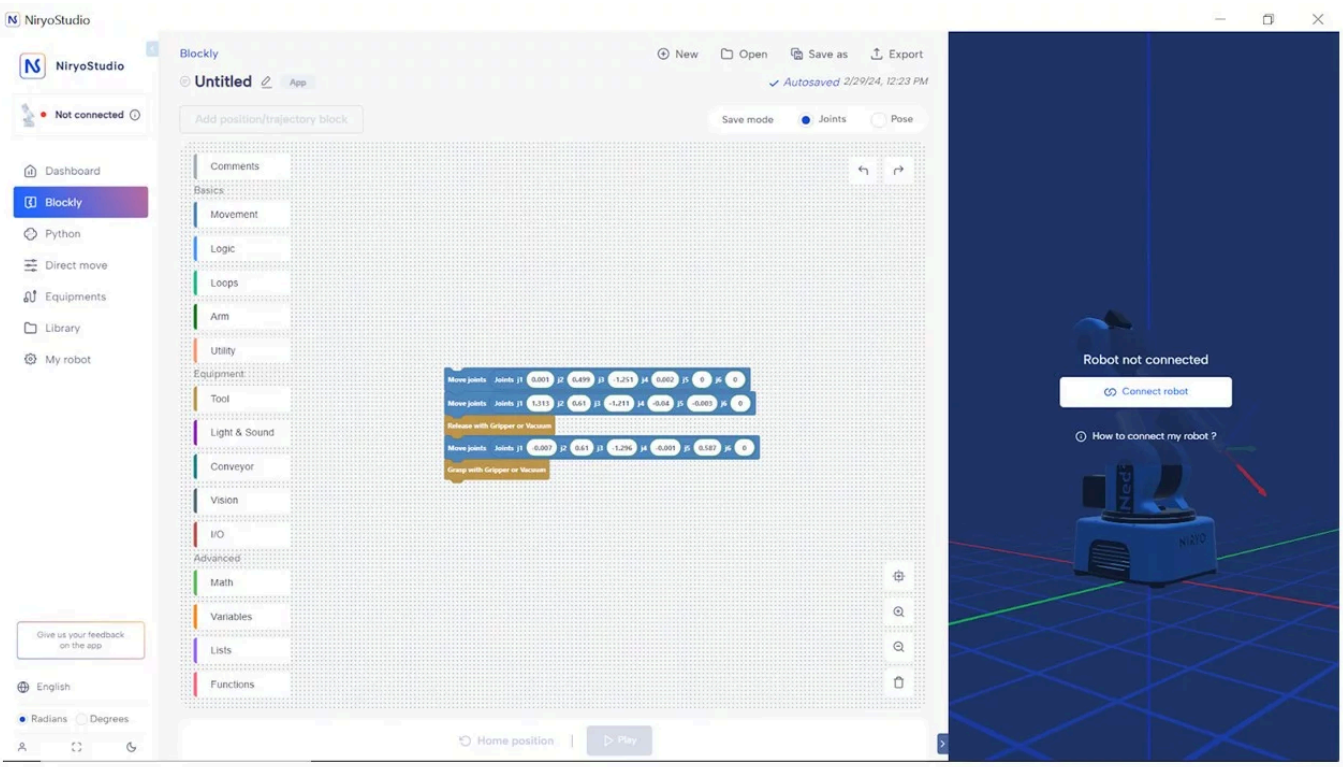
3. Chapter 3: Introduction to Blockly

Objective

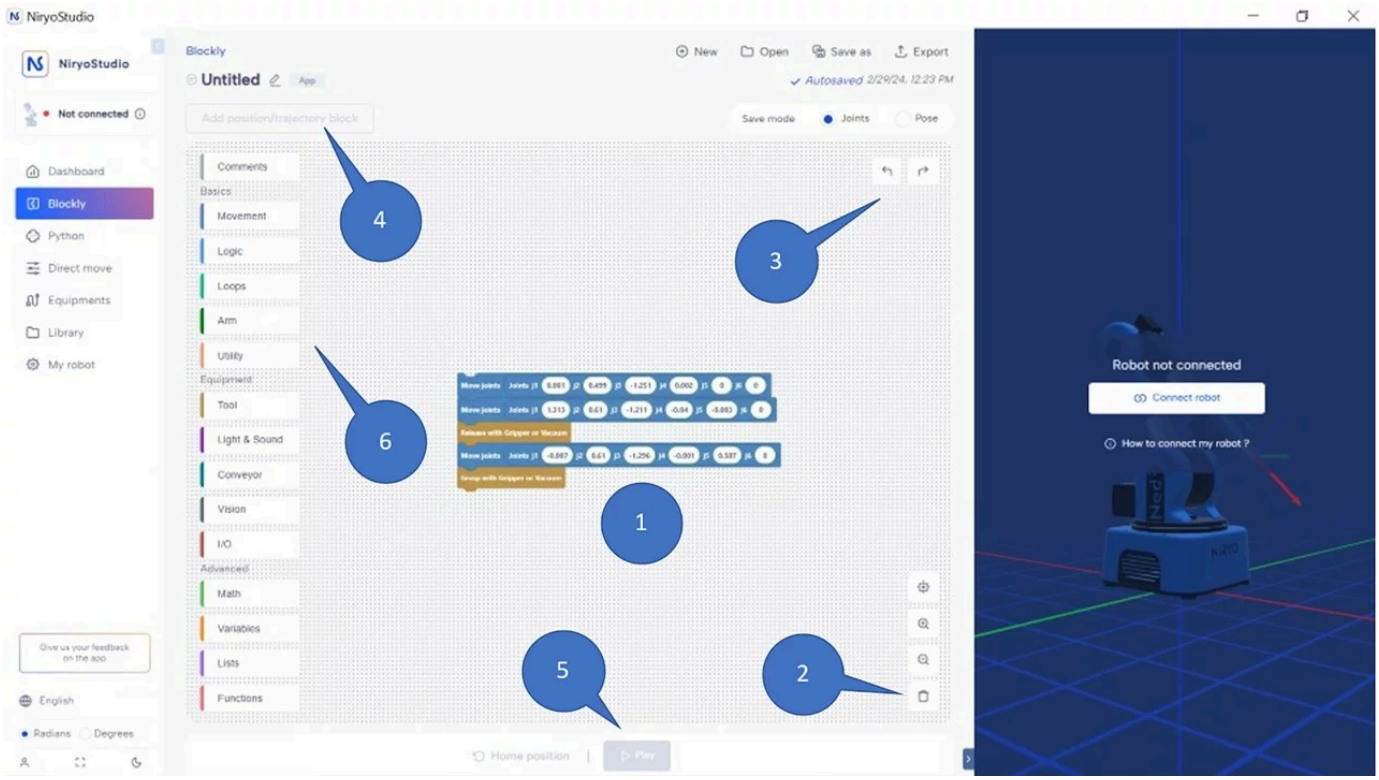
Getting started with Blockly

3.1. Step 1: Blockly Overview

NiryoStudio's visual interface is based on Blockly, an open source JavaScript library created by Google, which is also used for the Scratch project carried out by MIT. Blockly allows you to create development environments using a graphical language.

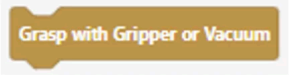
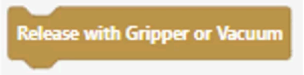


3.2. Step 2: The programming environment



1. Workspace which will contain the sequence that **the Ned2 robotic arm** will carry out.
2. Drag a block to the Trash a block to delete it.
3. Undo/Redo.
4. Add a position block or trajectory.
5. Execute the sequence displayed in the workspace.
6. Programming blocks.

3.3. Step 3: Blocks to use

Group	Name of Block	Explanation	Block
Comments	Comments	The “Comment” block allows you to add explanations to the sequence	
Basics, movement	Move Joints	The “move Joints” block moves the robot to the indicated position.	
Basics, utility	Wait for x seconds	The "Wait for x seconds" block stops the robot's movement for the desired time	
Equipment, tool	Grasp with Gripper or Vacuum	The "Grasp with Gripper or Vacuum" block closes the Gripper or sucks with the vacuum pump	
Equipment, tool	Release with Gripper or Vacuum	The "Release with Gripper or Vacuum" block opens the Gripper or releases the pressure from the vacuum pump	

4. Chapter 4: Programming with Blockly

Objective

Create the movements that the NED2 robotic arm must perform.

4.1. Step 1: Create the initial and final position of the NED2 robotic arm

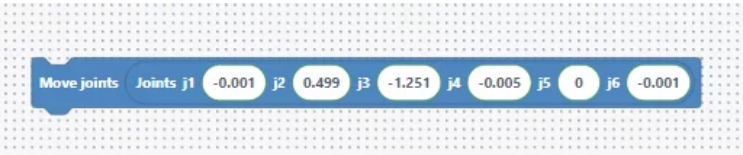
With NiryoStudio, create a home position.
To do this, press the “Home position” button.



And then short press the “Save” button on the control panel.



A position appears on NiryoStudio.



This position will be the first and last in the Blockly sequence.

4.2. Step 2: Create the sequence for taking the cup and positioning it on the coffee machine.

On the Ned2 robotic arm control panel, press and hold the “Freemotion” button. Move the NED2 robotic arm to pick up the coffee cup. To add positions, briefly press the “Save” button on the control panel. To open and close the gripper, briefly press the “Custom” button on the control panel.

> You must add “Grasp with gripper or vacuum pump” and “Release with gripper or vacuum pump” blocks. Pressing the "Custom" button does not automatically add Blockly blocks.



4.3. Step 3: Press the button on the coffee machine

Once the cup is placed on the coffee machine, open the grip then move the NED2 robotic arm away.
With precise movements, make the arm press the button of the coffee machine and wait for the cup to be filled. To do this, use the “WAIT for X Seconds” block.



Change the number of seconds depending on the speed of the coffee machine.

4.4. Step 4: Put the coffee on the table

Add the blocks needed to pick up and place the cup in its initial position and return the NED2 robotic arm to its initial position.

4.5. Final Solution

Solution

